

TICKET BOOKING WEB APPLICATION FOR TECHNICAL FEST

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering
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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "TICKET BOOKING WEB APPLICATION FOR TECHNICAL FEST" by AVVARU AISHWARYA(VTU26635) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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ABSTRACT

The Ticket Booking Application for a Technical Fest is a modern web-based platform designed to streamline the process of reserving tickets for institutional or organizational events. It enables users to browse event schedules, verify seat availability, and securely book tickets through an authenticated login interface. The system also provides an administrative dashboard where organizers can create events, manage bookings, and monitor participant data in real time. By digitizing the traditional booking process, the application minimizes manual errors, avoids double bookings, enhances data accuracy, and significantly reduces processing time. Overall, it improves operational efficiency while delivering a convenient and user-friendly experience for both participants and administrators.

Keywords: Keywords: Ticket Booking Application, Event Scheduling, Web Platform, Secure Authentication, Online Booking, Database Integration, Technical Fest, Automation

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CI	Continuous Integration
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
TX	Technical Fest Tickets
MVC	Model View Controller
ORM	Object Relational Mapping
TBDE	Ticket Booking Department Event
UI	User Interface
UX	User Experience

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Chapter 1

INTRODUCTION

1.1 Introduction

The Ticket Booking System for a Technical Fest is developed to offer a smooth, efficient, and dependable solution for handling event registrations within colleges or organizations. Conventional booking methods, such as manual entry or offline registration, often lead to delays, data inconsistencies, and difficulty in managing large numbers of participants. To overcome these challenges, this system provides an online platform where users can easily explore event information, verify ticket availability, and complete bookings through a secure login process. At the same time, administrators are given control to create and update events, track registrations, and manage participant records instantly. By automating the entire process, the system enhances accuracy, saves time, and ensures a better and more organized experience for both users and event coordinators.

1.2 Aim of the Project

The primary objective of this project is to design and implement an easy-to-use web-based application for booking tickets for Technical Fest events. The system aims to simplify and automate the entire registration process, making it faster and more efficient for users. By reducing dependency on manual methods, it minimizes errors, avoids duplication, and ensures accurate record management. Overall, the project focuses on enhancing user convenience while improving the efficiency and reliability of event registration.

1.3 Scope of the Project

The scope of the Ticket Booking System for an Technical Fest includes developing a web-based platform that allows users to view event details, check ticket availability, and book tickets online. It covers features such as secure user authentication, event creation, ticket management, and booking tracking by administrators. The system is intended for internal use within a department or organization, aiming to improve efficiency, reduce manual work, and ensure accurate and organized event management.

1.4 Framework

The system framework consists of layers like user interface, application logic, and database.

Figure 1.1 presents The system framework is organized into three distinct layers: the user interface, application logic, and database.

Figure 1.1: Framework

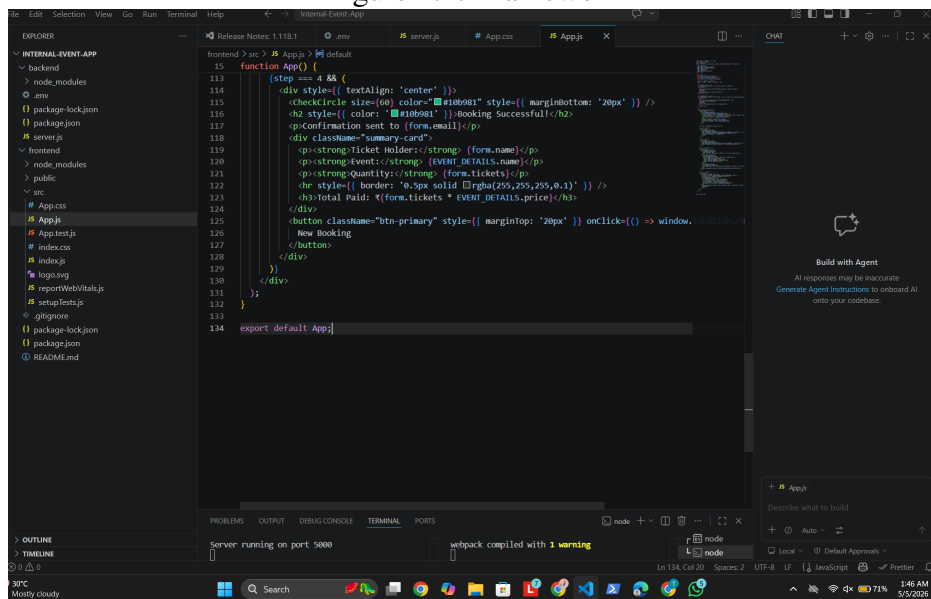


Figure 1.1 shows the visual illustrates this structure through a development environment showing the code and file hierarchy of a React application.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Several ticket booking and event management applications are widely used in the market, such as BookMyShow, Eventbrite, Ticketmaster, and Paytm Insider[1]. These platforms provide essential features including real-time ticket booking, secure online payment integration, event discovery, and user-friendly interfaces that enhance the overall user experience.

In addition to these, platforms like Eventzilla and Arlo Event Management offer advanced functionalities such as customizable event pages, attendee management, automated communication, and detailed reporting tools for organizers[2]. These features help in efficiently managing large-scale events and tracking participant engagement.

Furthermore, systems like Booking Ninjas provide real-time tracking, automated registration processes, and centralized dashboards, which

significantly improve operational efficiency and reduce manual effort[1]. Most existing applications are primarily designed for large-scale public events such as concerts, movies, and conferences. They commonly include features like seat selection, QR-based e-ticketing, multiple payment options, and analytics tools that support data-driven decision making.

However, these platforms often lack specific support for internal departmental or institutional events, where requirements such as restricted access, simplified booking, and internal communication are [2]. This gap highlights the need for a dedicated system tailored for internal event management and ticket booking.

Table 2.1: Comparative Analysis of Market Alternatives

Feature	BookMyShow	Eventbrite	Townscript	Your Project
Event Creation	Yes	Yes	Yes	Yes
Ticket Booking	Yes	Yes	Yes	Yes
Internal Department Events	No	No	No	Yes
User Registration/Login	Yes	Yes	Yes	Yes
Online Payment Integration	Yes	Yes	Yes	No / Planned
QR Code / E-Ticket	Yes	Yes	Yes	Yes
Seat Selection	Yes	Limited	Limited	Yes
Event Analytics	Yes	Yes	Yes	Basic
Direct Communication (Admin-User)	No	Limited	No	Yes

Table 2.1 presents displays a comparative analysis of alternatives.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend development environment is set up using modern web technologies such as HTML, CSS, and JavaScript. Tools like code editors (e.g., VS Code) and web browsers are used for development and testing. Required libraries and frameworks, if any, are installed to enhance design and functionality.

3.1.2 Structure of the Project

The project follows a modular structure to ensure scalability, maintainability, and clear separation of concerns:

/public: Contains static assets such as index.html, logos, event banners, and other images used across the application.

/src/api: Includes configuration files (e.g., axios.js) that define the base URL and handle communication with the backend server.

/src/components: Contains reusable UI components such as Navbar, Sidebar, Event Cards, Ticket Forms, and QR Code generator.

/src/context: Manages global state using context APIs (e.g., AuthContext.js) for user authentication, session handling, and role-based access (Admin/User).

/src/pages: Includes the main pages of the application-
Home.js: Displays available events and general information.
EventDetails.js: Shows detailed information about a selected event.
Booking.js: Handles ticket booking, seat selection, and user inputs.
Dashboard.js: Admin panel for managing events, bookings, and users.
MyTickets.js: Allows users to view and download their booked tickets.

/src/utlis: Contains helper functions such as validation, date formatting, and ticket generation logic.

App.js: The root component that defines routing, navigation, and overall layout of the application.

3.1.3 Execution Procedure

The project is executed by opening the main HTML file in a web browser or running it through a local development server[3]. Users can interact with the interface to view events, book tickets, and perform other actions, while developers can test and debug features during execution.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend ecosystem requires coordinated setup of multiple components to ensure smooth operation of the ticket booking and event management system:

Frontend Layer (React Application): The frontend is developed using React.js to create an interactive and user-friendly interface. Node.js (version 14 or above) is used to run the development server and manage the application environment[4]. Package managers such as npm or Yarn are utilized for installing and maintaining dependencies required for the project.

Backend Layer (Spring Boot): The backend is built using Spring Boot, which simplifies the development of RESTful APIs and business logic [5]. Java Development Kit (JDK) is required for compiling and running the application. Apache Maven is used for dependency management and handling the project build lifecycle efficiently.

Database Layer: The system uses MySQL (version 8.0 or above) as the primary database to store user details, event information, and booking records[3]. MySQL Workbench can be used optionally for database design, querying, and administration tasks.

Integration Tools: To ensure seamless communication between frontend and backend, Axios is used for making HTTP requests[1]. Postman is used for testing and validating APIs during development. Git is used for version control and collaborative development.

Deployment Environment (Optional): The application can be tested locally using a localhost server[2]. For production deployment, cloud platforms can be used to host the application, ensuring accessibility, scalability, and performance.

3.2.2 Structure of the Project

The backend project is organized in a modular structure with folders like controllers, services, and repositories. Each module handles specific tasks such as authentication, event management, and booking, making the system easy to maintain, scalable, and well-structured for future improvements.

3.2.3 Execution Procedure

The backend server is executed using appropriate commands such as `mvn spring-boot:run` or by running the packaged JAR file. Once the server is started, it listens for client requests from the frontend, processes the data, and interacts with the database to perform necessary operations. APIs are tested using tools like Postman to verify correctness and ensure proper communication between different components of the system.

Figure 3.1: Backend Implementation

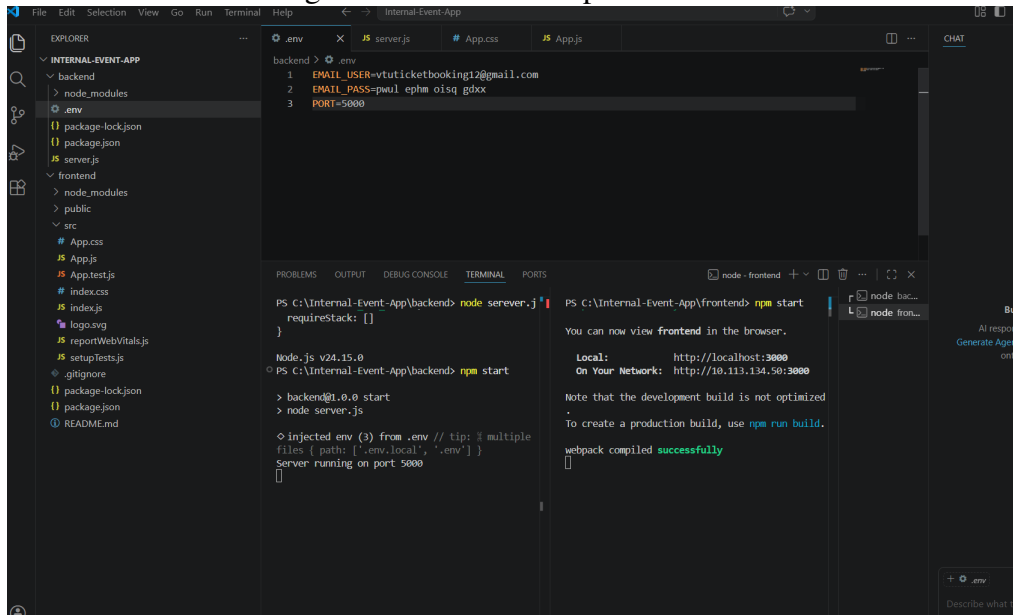


Figure 3.1 presents The backend of the Event Ticket Booking System is responsible for handling the application logic, user authentication, and communication with the database. It is developed using Node.js with the Express.js framework, which enables efficient handling of HTTP requests and server-side operations

3.3 Database Implementation

Database schema is created, followed by the creation of necessary tables such as Users, Events, and Bookings. Each table is designed with appropriate fields and primary keys to uniquely identify records. Relationships between tables are established using foreign keys to ensure data integrity.

3.3.1 Database Configuration

The system uses a relational database management system such as MySQL to organize and store application data. The database is properly set up by creating required schemas, defining tables, assigning user privileges, and establishing a secure connection with the backend. It efficiently manages and maintains records related to users, event details, and ticket bookings[5]. This structured approach ensures data consistency, quick retrieval, and secure storage, supporting smooth operation of the ticket booking application.

3.1.2 Schema Structure

The database schema is designed to organize data into structured tables with proper relationships[3]. It includes entities such as users, events, tickets, and bookings, connected through primary and foreign keys to maintain data integrity and consistency.

3.1.3 Tables created

The core functionalities are supported by the following primary tables:

Users: Stores user information including name, email, encrypted password, role (Admin/User), and contact details. It manages authentication and access control within the system.

Events: Contains details of all internal department events such as event name, description, date, time, venue, and organizer information.

Tickets: Stores ticket-related information including ticket ID, event ID, seat number (if applicable), and ticket type (general/VIP).

Bookings: Acts as a junction table between users and events, storing booking details such as booking ID, user ID, event ID, number of tickets, booking date, and status (Confirmed, Cancelled).

Payments: Maintains transaction records including payment ID, booking ID, payment method, amount, and payment status (Success, Failed, Pending).

3.4 System Technology Stack Overview

The following table provides a comprehensive summary of the frameworks and technologies utilized in the development of the Ticket Booking of an Internal Department Event system, categorized by their architectural role.

Table 3.1: System Technology Stack

Component	Technology
Frontend	React.js, HTML5, CSS3
Backend	Spring Boot, Spring Security
Database	MySQL
API Communication	Axios
Tools	Git, Postman, VS Code

Table 3.1 presents UI Interaction as the user clicks a button or submits a form in React frontend. API Request as React uses Axios to send HTTP Request to backend.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

Figure 4.1: Proposed System Architecture of Ticket Booking for an Internal Event Application

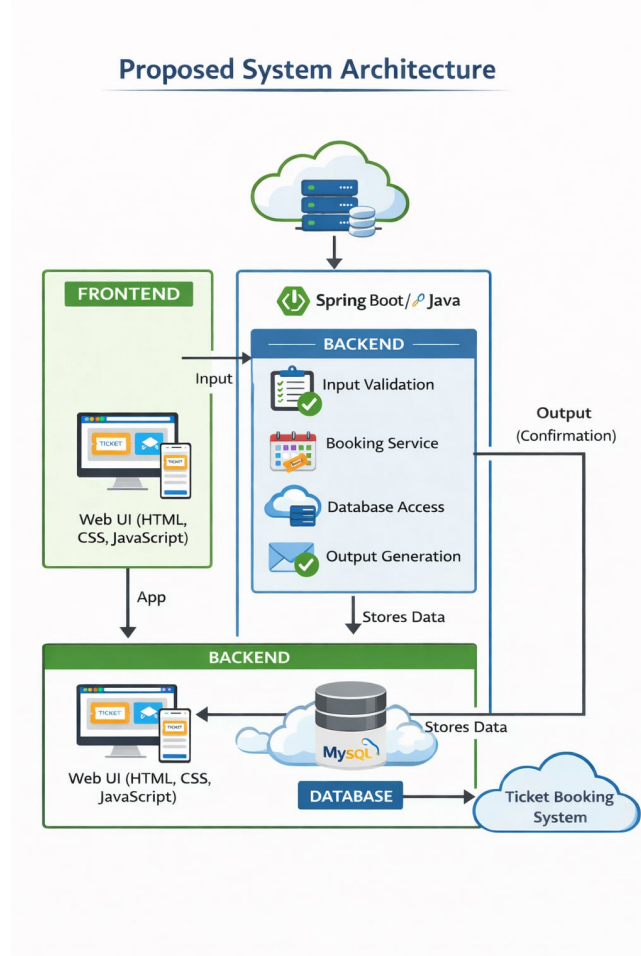


Figure 4.1 presents a full-stack architecture for a ticket booking application, utilizing a web-based frontend and a Java Spring Boot backend.

The project follows a layered architecture with frontend, backend, and database components. The frontend handles user interaction, the backend manages logic and processing, and the database stores all data. This structure ensures efficient, secure, and scalable system operation.

4.2 DataFlow Diagram

The Data Flow Diagram shows how data moves within the system. User inputs are processed by the backend and stored in the database. The system then returns the required output, such as booking confirmation, to the user.

4.3 Module 1:

User Management: This module handles user registration, login, and authentication. It ensures that only authorized users can access the system and perform booking operations.

It presents the workflow diagram illustrates the overall process of the internal event ticket booking system. It shows the sequence of steps a user follows to book a ticket for an event.

Figure 4.2: Data Flow Diagram of Ticket Booking for an Internal Event Application

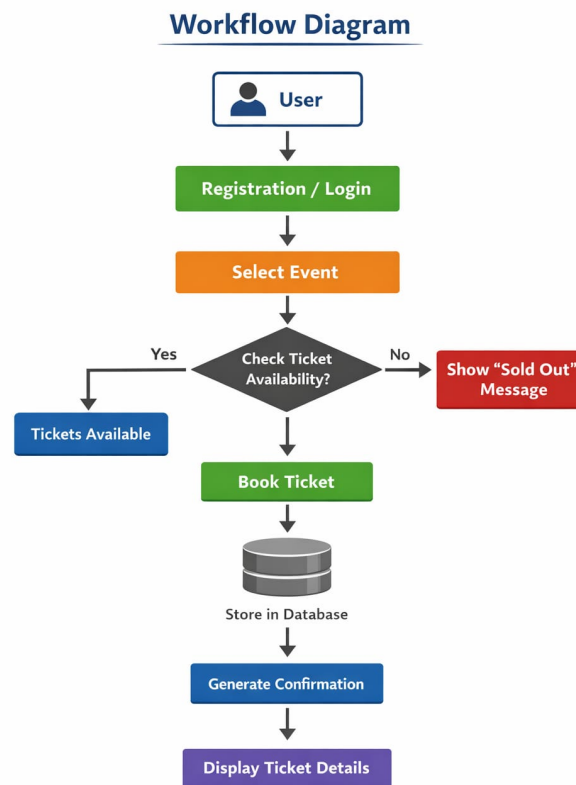


Figure 4.2 Accesses the system and completes the registration or login process for the events that are held in the college.

4.4 Module 2:

Event Management: This module allows administrators to create, update, and manage event details such as date, time, and ticket availability.

4.5 Module 3:

Ticket Booking: This module enables users to book tickets, check availability, and receive booking confirmation. It also prevents over-booking through proper validation.

Advantages of this Project The system simplifies ticket booking, reduces manual work, and improves efficiency. It ensures accurate data management and provides a user-friendly interface for both users and administrators.

a) Time Saving: Automates the booking process, reducing manual effort.

b) Error Reduction: Minimizes human errors and prevents over-booking.

c) Easy Management: Helps administrators manage events and bookings efficiently.

Disadvantages The system depends on internet connectivity and may face technical issues if not maintained properly. Initial setup and development may require time and technical knowledge.

Chapter 5

RESULTS AND DISCUSSIONS

5.1 Results and Discussion

Results: The Ticket Booking system for the technical fest was successfully developed and tested. The system efficiently handled user registration, event listing, ticket booking, and seat selection. Features such as QR code generation, payment processing, and user dashboard worked as expected. The admin dashboard enabled easy management of events, bookings, and users. Overall, the system provided a smooth and user-friendly experience with accurate data handling and real-time updates.

Screenshots:

Figure 5.1: Login Page Interface

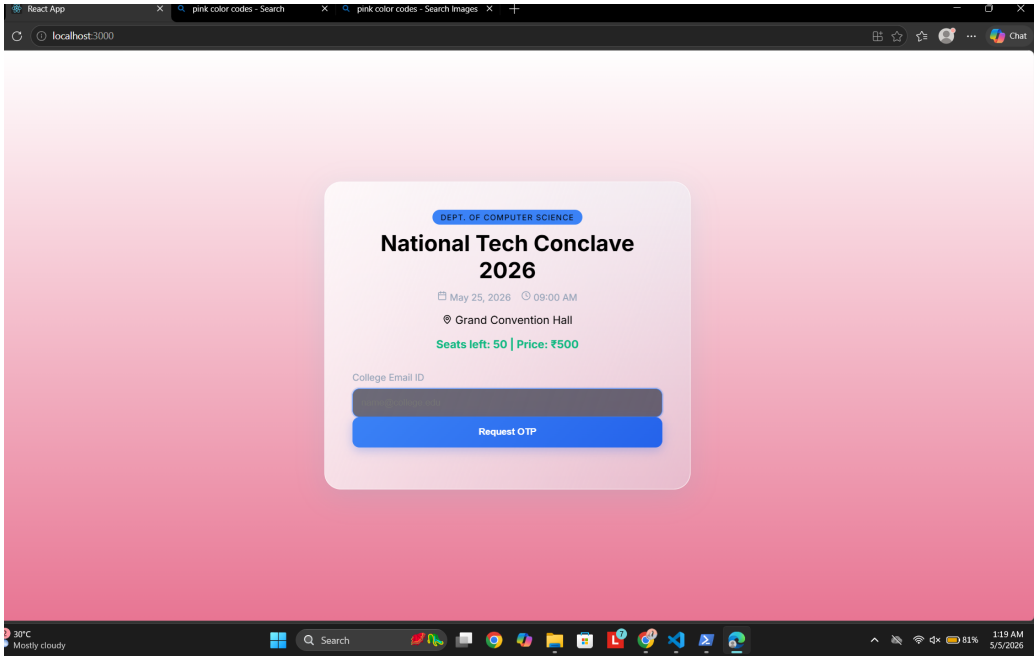


Figure 5.1 presents This document showcases the user interface design for the National Tech Conclave 2026 application.

Figure 5.2: Authentication Page

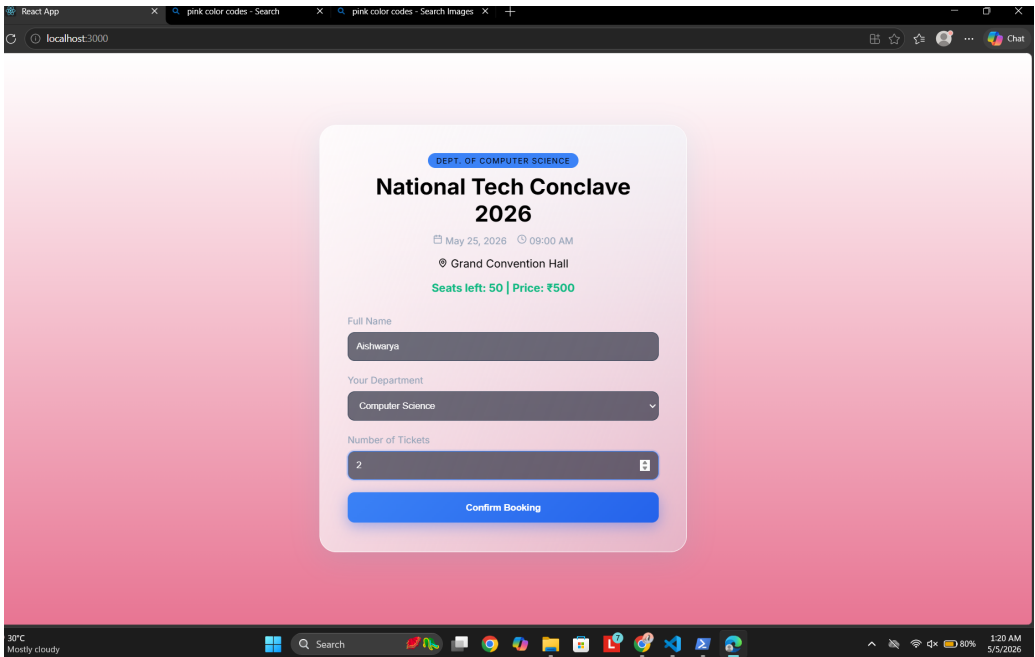


Figure 5.2 presents the authentication page is responsible for verifying the identity of users before allowing them to access the event booking system. This page ensures that only authorized users such as students or faculty members can use the application.

Figure 5.3: Otp Message

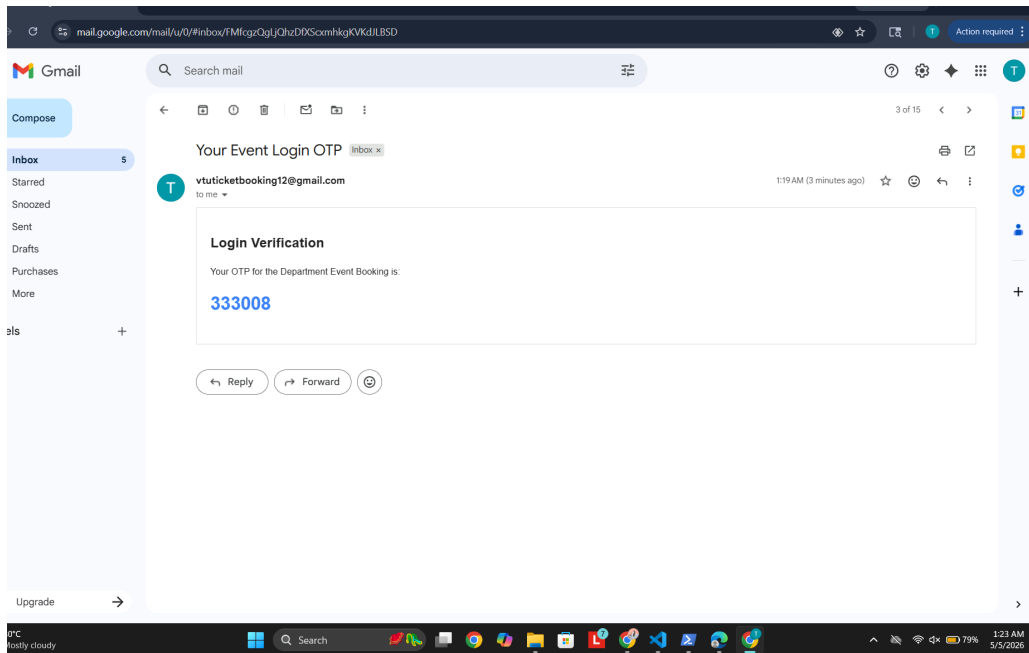


Figure 5.3 presents The user must check their email inbox and enter the received OTP in the verification field on the authentication page. This step ensures that the login request is made by the legitimate user who owns the email account.

Figure 5.4: Ticket Confirmation Page

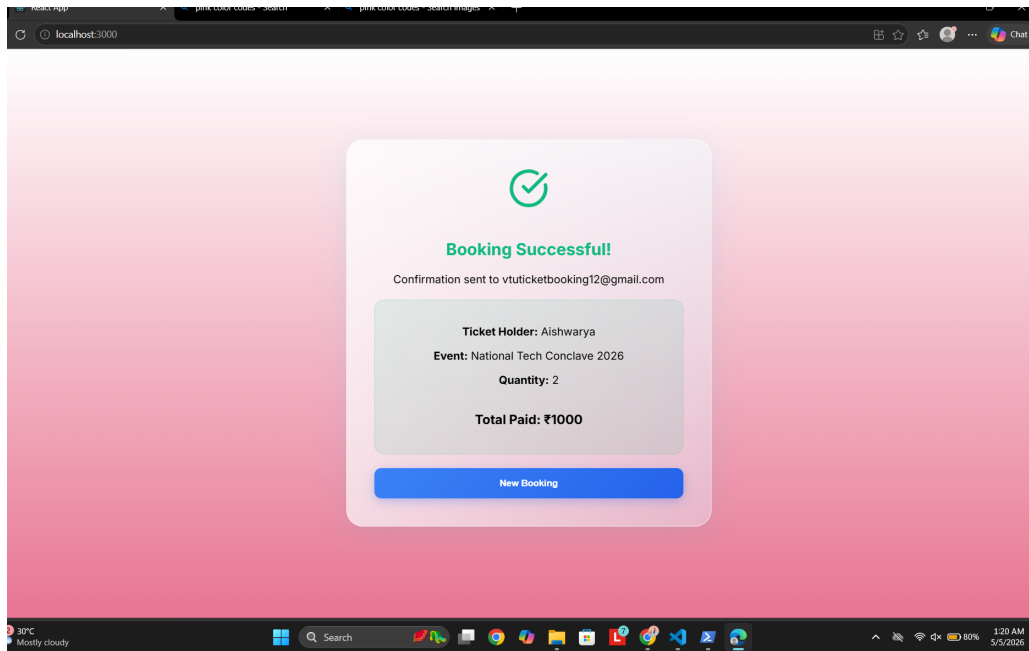


Figure 5.4 presents The ticket confirmation page is displayed after the user successfully completes the ticket booking process. This page confirms that the ticket has been booked and the booking details have been stored in the database.

Figure 5.5: Confirmation Mail Page

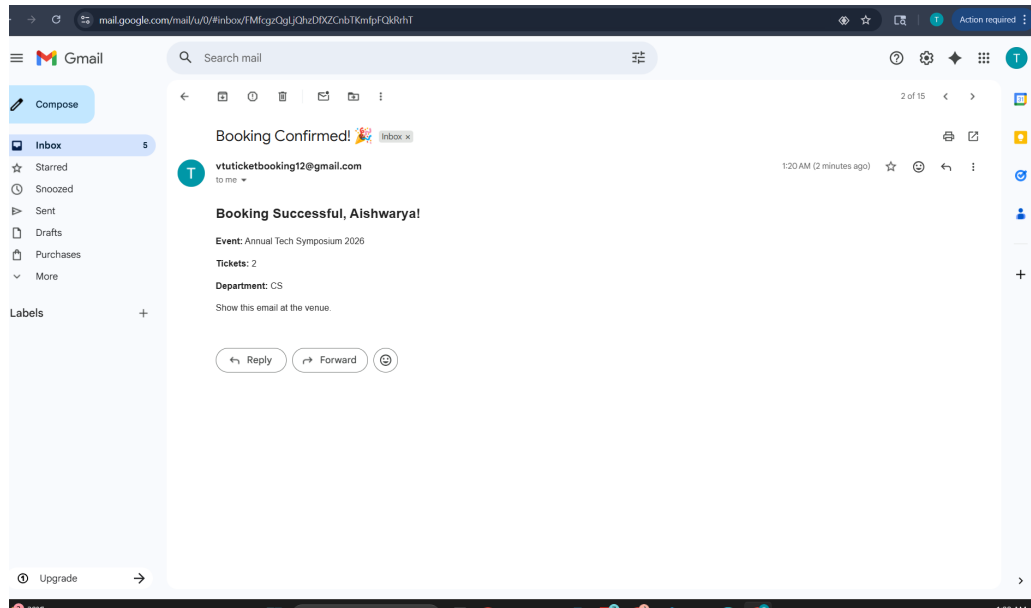


Figure 5.5 presents The confirmation email contains important details such as the event name, department name, event date and time, and booking confirmation information. This helps the user keep a record of their ticket booking.

Discussion: The Ticket Booking of an Internal Department Event system proves to be an effective solution for managing technical fest events[2]. Its modular architecture ensures scalability, security, and maintainability. The integration of multiple modules like authentication, booking, payment, and notification improves overall efficiency and reduces manual work. The system enhances coordination between users and administrators while ensuring reliable and organized event management.

Table 5.1: Ticket Booking System Module Comparison

Modules	Features
Landing Page	User-friendly homepage
Authentication	Login and registration
Event Listing	Displays event details
Ticket Booking	Books tickets easily
Seat Selection	Selects available seats
Payment	Handles online transactions
E-Ticket / QR Code	Generates digital tickets
User Dashboard	Shows booked tickets
Admin Dashboard	Manages events and bookings
Notifications	Sends confirmation alerts
Reports	Shows booking insights

Table 5.1 presents This table highlights the core modules and features of a Ticket Booking System, detailing essential functions like authentication, seat selection, and payment processing.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Ticket Booking System for an Internal Department Event successfully provides a simple and efficient solution for managing event registrations. It reduces manual work, minimizes errors, and ensures smooth ticket booking through a user-friendly interface. The system improves overall event management by providing better control, accuracy, and convenience for both users and administrators.

6.2 Future Enhancements

While the current implementation of the Ticket Booking of an Internal Department Event system provides a strong foundation for managing events and bookings, the following enhancements are planned for

future versions:

Online Payment Integration: Integration of secure payment gateways to enable seamless online transactions, supporting multiple payment options such as credit/debit cards and digital wallets.

QR Code-Based Entry System: Enhancing the ticket validation process by implementing QR code scanning at event entry points for faster and more secure check-ins.

Advanced Analytics Dashboard: Introducing detailed analytics for administrators, including attendee insights, booking trends, and event performance metrics for better decision-making.

Email and SMS Notification System: Automated alerts for booking confirmations, event reminders, cancellations, and updates to improve user engagement.

Role-Based Access Control Enhancement: Expanding user roles (e.g., Super Admin, Organizer, Staff) to improve system security and provide better control over event management operations.

Seat Allocation Optimization: Implementing intelligent seat allocation algorithms to optimize seating arrangements and enhance user convenience.

Chapter 7

Addressing Complex Engineering Problem and SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

WP1	Depth of Knowledge	Requires in-depth engineering	WK4, WK5	Knowledge of React JS, hooks, validation, UI design
WP2	Conflicting Requirements	Technical and non-technical constraints	WK4, WK5	Balances usability with ticket booking validation
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4	Component design, state handling, dynamic updates
WP4	Familiarity	Partially new problems	WK4	Real-time booking and ticket updates
WP5	Applicable Codes	Limited standard solutions	WK6	Custom booking and validation logic
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6	Students, faculty, organizers
WP7	Interdependence	System-level integration	WK3, WK5	UI + state + validation integration

Figure 7.1 presents the table presents the mapping of the Event Ticket Booking System project to the Complex Engineering Problem (CEP) characteristics. It explains how different aspects of the project correspond to various problem attributes such as depth of knowledge, analysis, stakeholder needs, and system integration.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Problem (CEP)

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources	Use of technologies	React JS, HTML, CSS
EA2	Level of Interactions	Complex module interaction	Event display + booking + validation
EA3	Innovation	Modern technologies	Hooks, conditional rendering
EA4	Consequences	Impact on users	Simplifies booking process
EA5	Familiarity	Beyond standard solutions	Dynamic real-time system

Figure 7.2 presents the table illustrates how the Event Ticket Booking System project is mapped to various Complex Engineering Activities (CEA) attributes. It highlights the technologies used, the level of interaction between system modules, and the innovative aspects involved in the development process.

7.3 Sustainable Development Goals (SDG) Mapping

Table 7.3: SDG Mapping of the Project

SDG No.	SDG Title	Relevance
SDG 4	Quality Education	Supports academic events and learning
SDG 9	Industry, Innovation and Infrastructure	Promotes modern React-based systems
SDG 10	Reduced Inequalities	Accessible platform for all users
SDG 11	Sustainable Cities and Communities	Encourages organized and efficient event management

Figure 7.3 presents the table illustrates how the Event Ticket Booking System project is mapped to various Complex Engineering Activities (CEA) attributes. It highlights the technologies used, the level of interaction between system modules, and the innovative aspects involved in the development process.

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